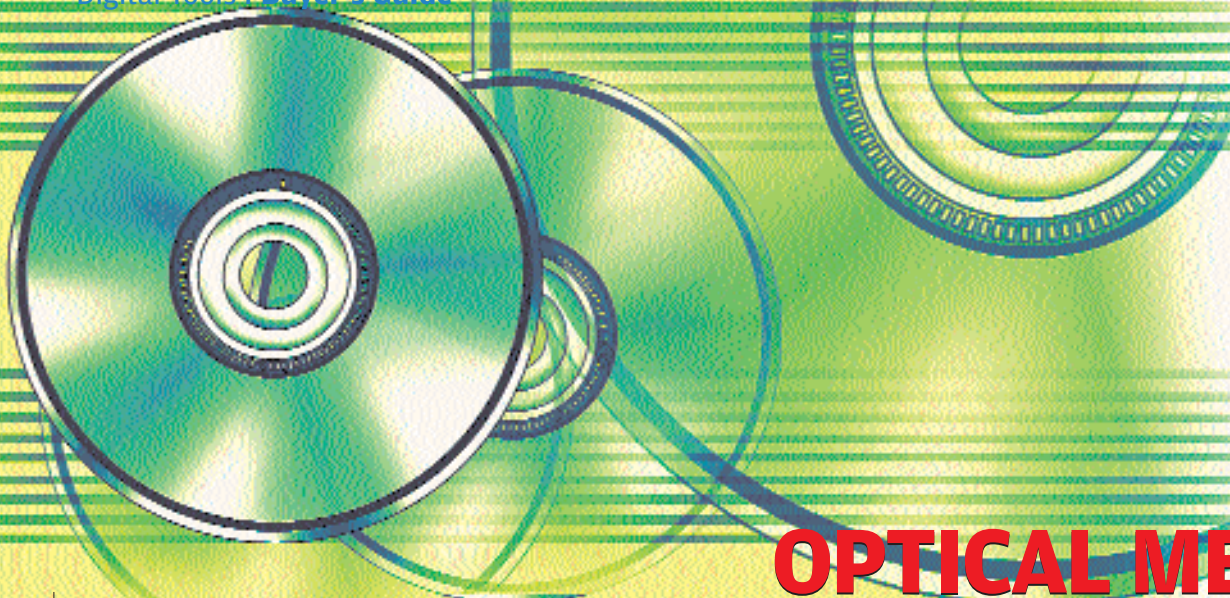




OPTICAL MEDIA

Computers rely on optical storage media such as CDs and DVDs to back up and store data because of the sheer speed, the huge storage capacities, and also convenience and cost. The booming video CD, audio CD and movie DVD industries are testimony to the success of the optical storage medium. As important as the optical media writers may be, the optical storage media is no less important.

Questions to ask the dealer

Is the media compatible with all optical drives?

Some brands of CD and DVD media have known compatibility problems with certain brands of CD and DVD drives. Some encounter problems burning in a specified brand of optical writer, whereas other media can be burnt perfectly in the same writer. Such problems are usually specified on the drive and/or media manufacturer's Web site, and before purchasing the media, you should do some research about whether it will work in your optical drive.

Is it certified to burn at a specified speed?

Every optical media is certified to be able to be written to at a certain maximum

speed. Check this, and buy the one that supports the highest speed.

Is it scratch-resistant?

There are some media available which have a protective coating that reduces its susceptibility to getting scratched. Though such media come at a premium, go for them if the data you are writing is very important.

What is the maximum capacity of the disc?

There are some discs available whose sizes are a cut above the regular media, for example, 800 MB CDs. These can be useful if you have files whose sizes exceed 700 MB. Again, your CD/DVD writer should support writing to such a disc.

What is the warranty, if any?

Some manufacturers provide a replacement warranty for their media, and you should check which manufacturers do so.

Usage tips

While labelling on the CD/DVD media face using a marker pen, use the lowest possible pressure so as to avoid damage.

If burning at the maximum speed almost always produces a coaster, try lowering the speed.

Avoid touching the surface of the disc.

While cleaning a CD/DVD, use a soft

lint cloth. Do not clean the CD/DVD in a circular motion—clean across the radius, moving from the centre towards the edge in a straight line.

Store optical media in a cool place away from direct sunlight and other sources of heat, as this is likely to damage the medium.



While creating an audio CD, burn it at a slow speed, such as 8x, so that older CD players with poorer lasers will be able to play the disc thus created.

Do not rely solely on optical media for data storage and backup, as they tend to degrade over a period of time and become unreliable.

The Future

Optical media have evolved from CD to DVD and now to Blu-ray and HD-DVD. This has been possible with technological advances that make it possible to use lasers of higher frequency and shorter wavelength, which makes it possible to store a larger amount of data on a disc of the same size.

Blu-ray, with storage capacities of 25 or 50 GB, and HD-DVD, with capacities of 15, 30 or 45 GB, are almost here, and are set to replace DVD. These make it possible to record over two hours of HDTV—or, for us, over 13 hours of standard TV broadcast. Some way in the future is 3D Volume Optical Storage Nanotechnology, which promises to be able to cram 100 terabytes of data onto a single disc! ■



Points To Ponder

Will the particular type of media work in my drive?

This problem is regarding DVDs. There are many different types of DVDs, including DVD-R, DVD+R, DVD-RW, DVD+RW, DVD-R DL, DVD+R DL and DVD-RAM. Most drives support almost every type of media these days, but there are some media such as DVD-RAM which very few drives support. Also, some drives may read dual-layer DVDs, but may support writing only to DVD+R dual-layer media. Also find out if the specified media capacity (such as 800 MB CDs) is supported by your optical drive.

Are there compatibility problems with various DVD formats?

There are certain incompatibilities in certain types of DVD media. As an example, a DVD+R/RW drive can't write a DVD-R or DVD-RW disc, and vice versa.

Does it support overburning?

Overburning is the process of writing to an optical disc beyond its specified capacity. As an example, using overburning, 715 MB of data may be written to a 700 MB disc. This has to be supported by the media as well as the drive, and also by the CD/DVD writing software.